

Spatial Boundary Modeling & GET for the 1960s

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Overview

In this R markdown document, we outline the procedure of modeling the primary boundary describe in “A Statistical Framework for Spatial Boundary Estimation and Change Detection: Application to the Sahel-Sahara Climate Transition”. We first begin by downloading the appropriate packages to model and apply the GET for the primary boundary.

```
# Set global chunk options
knitr::opts_chunk$set(echo = TRUE, warning = FALSE,
  message = FALSE, fig.width = 8, fig.height = 5)

# Load required libraries
library(dplyr)
library(hetGP)
library(splines)
library(ggplot2)
library(MASS)
library(sf)
library(rnaturalearth)
library(rnaturalearthdata)

set.seed(1)
```

Data Collection

We collect the primary and secondary boundary lines using the `get_data` function and temporal radial basis functions are evaluated to capture the temporal component of the boundary. Six different periods and four different radial functions are created to make 24 unique temporal functions applied over the time window. The primary boundary points are selected from the CLASS: “Des_Line”, and is defined between a specified longitude and latitude coordinates.

```
# Helper function to load and sort yearly
# data
get_data <- function(year) {
  # Fixed the duplicate read.csv line
  data <- read.csv(paste0("Two_Lines", year,
    "_dup.csv"))
  data$year <- as.numeric(year)
  df_sorted <- data[order(data$lon), ]
}
```

```

    return(df_sorted)
}

a <- 1
time <- 1960:1989

# FIX: Wrap scale() in as.vector() to keep
# it a simple numeric vector
time_scaled <- as.vector(scale(time, center = TRUE,
    scale = FALSE))

# Define periodicities
T_period1 <- 18
T_period2 <- 3
T_period <- 6
T_period3 <- 9
T_period4 <- 12
T_period5 <- 15

rbf_basis <- cbind(a * sin(2 * pi * 1 * time_scaled/T_period),
    a * cos(2 * pi * 1 * time_scaled/T_period),
    a * sin(2 * pi * 2 * time_scaled/T_period),
    a * cos(2 * pi * 2 * time_scaled/T_period),
    a * sin(2 * pi * 1 * time_scaled/T_period1),
    a * cos(2 * pi * 1 * time_scaled/T_period1),
    a * sin(2 * pi * 2 * time_scaled/T_period1),
    a * cos(2 * pi * 2 * time_scaled/T_period1),
    a * sin(2 * pi * 1 * time_scaled/T_period2),
    a * cos(2 * pi * 1 * time_scaled/T_period2),
    a * sin(2 * pi * 2 * time_scaled/T_period2),
    a * cos(2 * pi * 2 * time_scaled/T_period2),
    a * sin(2 * pi * 1 * time_scaled/T_period3),
    a * cos(2 * pi * 1 * time_scaled/T_period3),
    a * sin(2 * pi * 2 * time_scaled/T_period3),
    a * cos(2 * pi * 2 * time_scaled/T_period3),
    a * sin(2 * pi * 1 * time_scaled/T_period4),
    a * cos(2 * pi * 1 * time_scaled/T_period4),
    a * sin(2 * pi * 2 * time_scaled/T_period4),
    a * cos(2 * pi * 2 * time_scaled/T_period4),
    a * sin(2 * pi * 1 * time_scaled/T_period5),
    a * cos(2 * pi * 1 * time_scaled/T_period5),
    a * sin(2 * pi * 2 * time_scaled/T_period5),
    a * cos(2 * pi * 2 * time_scaled/T_period5))

data <- get_data("1960")
data <- cbind(data, t1 = rbf_basis[1, 1], t2 = rbf_basis[1,
    2], t3 = rbf_basis[1, 3], t4 = rbf_basis[1,
    4], t5 = rbf_basis[1, 5], t6 = rbf_basis[1,
    6], t7 = rbf_basis[1, 7], t8 = rbf_basis[1,
    8], t9 = rbf_basis[1, 9], t10 = rbf_basis[1,
    10], t11 = rbf_basis[1, 11], t12 = rbf_basis[1,
    12], t13 = rbf_basis[1, 13], t14 = rbf_basis[1,
    14], t15 = rbf_basis[1, 15], t16 = rbf_basis[1,

```

```

16], t17 = rbf_basis[1, 17], t18 = rbf_basis[1,
18], t19 = rbf_basis[1, 19], t20 = rbf_basis[1,
20], t21 = rbf_basis[1, 21], t22 = rbf_basis[1,
22], t23 = rbf_basis[1, 23], t24 = rbf_basis[1,
24])

# Merge subsequent years
for (j in 1961:1989) {
  data2 <- get_data(as.character(j))
  data2$year <- as.numeric(j)
  p <- as.numeric(j - 1970 + 11)

  data2 <- cbind(data2, t1 = rbf_basis[p, 1],
    t2 = rbf_basis[p, 2], t3 = rbf_basis[p,
    3], t4 = rbf_basis[p, 4], t5 = rbf_basis[p,
    5], t6 = rbf_basis[p, 6], t7 = rbf_basis[p,
    7], t8 = rbf_basis[p, 8], t9 = rbf_basis[p,
    9], t10 = rbf_basis[p, 10], t11 = rbf_basis[p,
    11], t12 = rbf_basis[p, 12], t13 = rbf_basis[p,
    13], t14 = rbf_basis[p, 14], t15 = rbf_basis[p,
    15], t16 = rbf_basis[p, 16], t17 = rbf_basis[p,
    17], t18 = rbf_basis[p, 18], t19 = rbf_basis[p,
    19], t20 = rbf_basis[p, 20], t21 = rbf_basis[p,
    21], t22 = rbf_basis[p, 22], t23 = rbf_basis[p,
    23], t24 = rbf_basis[p, 24])

  data <- data %>%
    full_join(data2)
}

# Subsetting to region of interest
des_data1 <- subset(data, CLASS == "Des_Line" &
  lon < 30)
des_data2 <- subset(data, CLASS == "Des_Line" &
  lon > 30 & lat > 9)
des_data <- rbind(des_data2, des_data1)

```

Temporal Decade Boundary Modeling

Latitude values from the selected boundary are regressed on by the radial temporal basis values, removing the temporal fluctuation of the line. The predicted latitude means for each of the years in the decades: 1960s, 1970s, and 1980s are averaged over to find average temporal mean for each decade.

```

# Regress latitude on radial basis function
# values
time <- lm(lat ~ t1 + t2 + t3 + t4 + t5 + t6 +
  t7 + t8 + t9 + t10 + t11 + t12 + t13 + t14 +
  t15 + t16 + t17 + t18 + t19 + t20 + t21 +
  t22 + t23 + t24, data = des_data)

# EXACT mean fitted values retained
Decade_1960 <- mean(c(13.07086, 13.54703, 13.41153,

```

```

13.33068, 13.29019, 13.04238, 13.30107, 13.71846,
13.55186, 13.3137))
Decade_1970 <- mean(c(12.97878, 12.38073, 12.34938,
12.67225, 12.70774, 12.92903, 13.10492, 12.83255,
12.84458, 12.99196))
Decade_1980 <- mean(c(12.74365, 12.65005, 12.54267,
12.08559, 12.041, 12.23725, 12.14689, 12.41581,
12.98462, 13.48154))

```

Spatial Boundary Modeling

The remaining spatial residuals are modeled using the `mleHetGP` function where the covaries are scaled longitude values. The training set of the Gaussian process is divided into three different training sets the 1960s, 1970s, and 1980s. One the decade boundary is train over its' respective data, spatial predictions are made over a fixed prediction grid, where the average temporal mean is added to the spatial prediction. Therefore we can construct the mean decade boundary line of the 1960s, 1970s and 1980s (`prediction_1960`, `prediction_1970`, `prediction_1980`).

```

# Standardize covariates
Y <- as.vector(time$residuals)
X1 <- (as.vector(des_data["lon"])$lon - min(as.vector(des_data["lon"])$lon))/(max(as.vector(des_data["lon"])$lon) - min(as.vector(des_data["lon"])$lon))

X01 <- data.frame(X = as.vector(des_data["lon"])$lon,
  Scaled_X = X1, YR = des_data$year, Lat = Y)

# Subdivide training years by decades
train <- subset(X01, YR < 1970 & YR >= 1960)
train2 <- subset(X01, YR < 1980 & YR >= 1970)
train3 <- subset(X01, YR < 1990 & YR >= 1980)

# Fit spatial_models
spatial_model_1960 <- mleHetGP(X = as.matrix(train["Scaled_X"]),
  Z = as.matrix(train["Lat"]), covtype = "Matern3_2")
spatial_model_1970 <- mleHetGP(X = as.matrix(train2["Scaled_X"]),
  Z = as.matrix(train2["Lat"]), covtype = "Matern3_2")
spatial_model_1980 <- mleHetGP(X = as.matrix(train3["Scaled_X"]),
  Z = as.matrix(train3["Lat"]), covtype = "Matern3_2")

# Generate a fixed set grid to make
# predictions
scale_pred <- seq(0, 1, length = 1000)
real_values <- seq(min(X01$X), max(X01$X), length = 1000)
test_grid <- cbind(scale_pred, real_values)
x <- test_grid[, 2]

spatial_1960 <- predict(object = spatial_model_1960,
  x = as.matrix(test_grid[, 1]))
spatial_1970 <- predict(object = spatial_model_1970,
  x = as.matrix(test_grid[, 1]))

```

```

spatial_1980 <- predict(object = spatial_model_1980,
  x = as.matrix(test_grid[, 1]))

prediction_1960 <- Decade_1960 + spatial_1960$mean
prediction_1970 <- Decade_1970 + spatial_1970$mean
prediction_1980 <- Decade_1980 + spatial_1980$mean

```

Visualization of Decade Boundary Line

Below are two plots of the predicted primary boundary line for each decade, with the associated prediction intervals as well. For each prediction boundary location in the respective decade there is an associated process variance (`spatial_1960$sd2`, `spatial_1970$sd2`, `spatial_1980$sd2`) and nugget (`spatial_1960$nug`, `spatial_1970$nug`, `spatial_1980$nug`) variance that composes the total variance. In this way the spatial model takes into account global and local variation in the entire boundary. To display the boundary lines and prediction intervals at a proper scale, a template of the continent Africa is used as background reference. Note that the mean boundaries and the predicted confidence intervals overlap each other with little deviation across the decades. One clear distinction between the decades is 1980s has larger predicted interval close to the horn region and the boundary shifts to the south slightly for the three decades.

```

# Get Africa polygon
africa <- ne_countries(continent = "Africa", returnclass = "sf")

# Prediction intervals
lower_sd2 <- qnorm(0.05, prediction_1960, sqrt(spatial_1960$sd2))
upper_sd2 <- qnorm(0.95, prediction_1960, sqrt(spatial_1960$sd2))
lower_sd2_nugs <- qnorm(0.05, prediction_1960,
  sqrt(spatial_1960$sd2 + spatial_1960$nugs))
upper_sd2_nugs <- qnorm(0.95, prediction_1960,
  sqrt(spatial_1960$sd2 + spatial_1960$nugs))

lower_sd2_2 <- qnorm(0.05, prediction_1970, sqrt(spatial_1970$sd2))
upper_sd2_2 <- qnorm(0.95, prediction_1970, sqrt(spatial_1970$sd2))
lower_sd2_nugs_2 <- qnorm(0.05, prediction_1970,
  sqrt(spatial_1970$sd2 + spatial_1970$nugs))
upper_sd2_nugs_2 <- qnorm(0.95, prediction_1970,
  sqrt(spatial_1970$sd2 + spatial_1970$nugs))

lower_sd2_3 <- qnorm(0.05, prediction_1980, sqrt(spatial_1980$sd2))
upper_sd2_3 <- qnorm(0.95, prediction_1980, sqrt(spatial_1980$sd2))
lower_sd2_nugs_3 <- qnorm(0.05, prediction_1980,
  sqrt(spatial_1980$sd2 + spatial_1980$nugs))
upper_sd2_nugs_3 <- qnorm(0.95, prediction_1980,
  sqrt(spatial_1980$sd2 + spatial_1980$nugs))

# Combine into data frames
lines_df <- rbind(data.frame(lon = x, lat = prediction_1960,
  group = "Mean", category = "Mean"), data.frame(lon = x,
  lat = lower_sd2, group = "Lower (sd2)", category = "Confidence"),
  data.frame(lon = x, lat = upper_sd2, group = "Upper (sd2)",
  category = "Confidence"), data.frame(lon = x,

```

```

    lat = lower_sd2_nugs, group = "Lower (sd2+nugs)",
    category = "Prediction"), data.frame(lon = x,
    lat = upper_sd2_nugs, group = "Upper (sd2+nugs)",
    category = "Prediction"))

lines_df_2 <- rbind(data.frame(lon = x, lat = prediction_1970,
    group = "Mean", category = "Mean"), data.frame(lon = x,
    lat = lower_sd2_2, group = "Lower (sd2)",
    category = "Confidence"), data.frame(lon = x,
    lat = upper_sd2_2, group = "Upper (sd2)",
    category = "Confidence"), data.frame(lon = x,
    lat = lower_sd2_nugs_2, group = "Lower (sd2+nugs)",
    category = "Prediction"), data.frame(lon = x,
    lat = upper_sd2_nugs_2, group = "Upper (sd2+nugs)",
    category = "Prediction"))

lines_df_3 <- rbind(data.frame(lon = x, lat = prediction_1980,
    group = "Mean", category = "Mean"), data.frame(lon = x,
    lat = lower_sd2_3, group = "Lower (sd2)",
    category = "Confidence"), data.frame(lon = x,
    lat = upper_sd2_3, group = "Upper (sd2)",
    category = "Confidence"), data.frame(lon = x,
    lat = lower_sd2_nugs_3, group = "Lower (sd2+nugs)",
    category = "Prediction"), data.frame(lon = x,
    lat = upper_sd2_nugs_3, group = "Upper (sd2+nugs)",
    category = "Prediction"))

uncertainty_df <- rbind(data.frame(lon = x, lat = lower_sd2_nugs,
    Decade = "1960s", Bound = "Lower"), data.frame(lon = x,
    lat = upper_sd2_nugs, Decade = "1960s", Bound = "Upper"),
    data.frame(lon = x, lat = lower_sd2_nugs_2,
    Decade = "1970s", Bound = "Lower"), data.frame(lon = x,
    lat = upper_sd2_nugs_2, Decade = "1970s",
    Bound = "Upper"), data.frame(lon = x,
    lat = lower_sd2_nugs_3, Decade = "1980s",
    Bound = "Lower"), data.frame(lon = x,
    lat = upper_sd2_nugs_3, Decade = "1980s",
    Bound = "Upper"))

# Get Africa polygon
africa <- ne_countries(continent = "Africa", returnclass = "sf")

# Plot mean lines
mean_lines <- rbind(data.frame(lon = x, lat = prediction_1960,
    line = "1960s"), data.frame(lon = x, lat = prediction_1970,
    line = "1970s"), data.frame(lon = x, lat = prediction_1980,
    line = "1980s"))

p <- ggplot() + geom_sf(data = africa, fill = "lightgray",
    color = "black") + geom_line(data = mean_lines,
    aes(x = lon, y = lat, color = line), size = 0.6) +
    scale_color_manual(values = c(`1980s` = "blue",
    `1970s` = "red", `1960s` = "green")) +

```

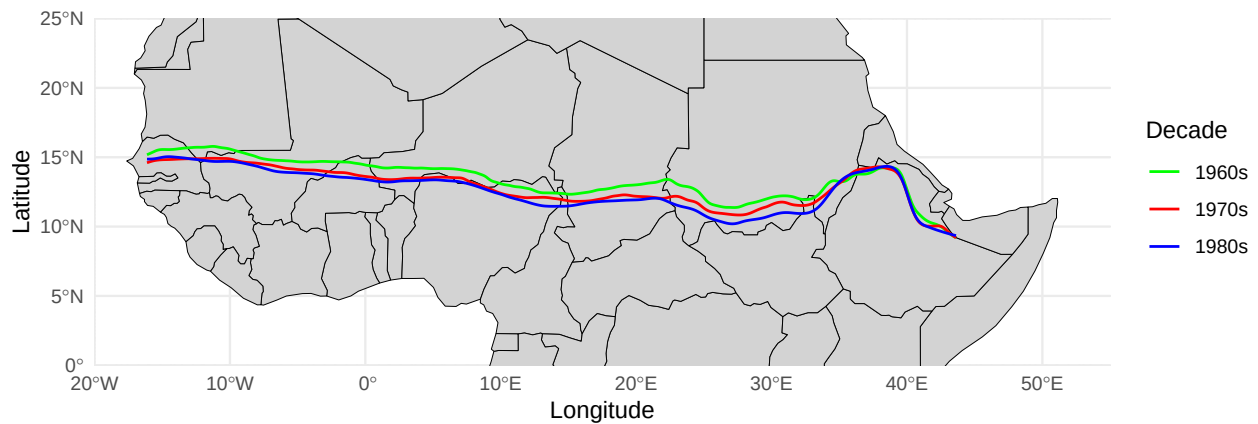
```

coord_sf(xlim = c(-20, 55), ylim = c(0, 25),
  expand = FALSE) + theme_minimal() + theme(plot.margin = margin(0,
0, 0, 0)) + labs(x = "Longitude", y = "Latitude",
color = "Decade")

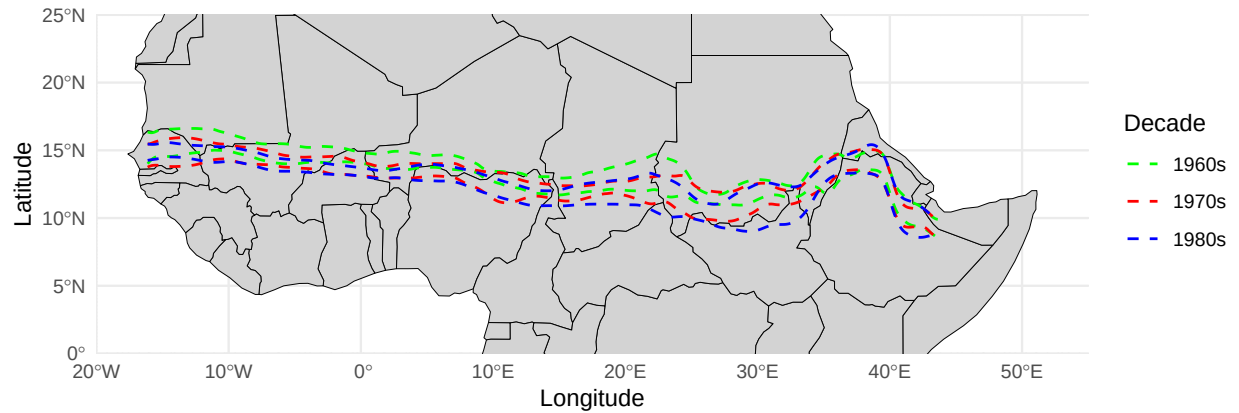
q <- ggplot() + geom_sf(data = africa, fill = "lightgray",
color = "black") + geom_line(data = uncertainty_df,
aes(x = lon, y = lat, color = Decade, group = interaction(Decade,
Bound)), linewidth = 0.6, linetype = "dashed") +
scale_color_manual(values = c(`1960s` = "green",
`1970s` = "red", `1980s` = "blue")) +
coord_sf(xlim = c(-20, 55), ylim = c(0, 25),
expand = FALSE) + theme_minimal() + labs(x = "Longitude",
y = "Latitude", color = "Decade")

print(p)

```



```
print(q)
```



Decade Global Envelope Test (GET)

We implement the GET for the decade boundary of the 1960s to see which yearly boundary lines in the 1970s and 1980s deviate from the 1960's global envelope. Below we construct the GET by generate the variance estimated from the 1960s line to see if the points fall within the the GET or not.

```
# Simulations and Envelope Construction
Kx <- cov_gen(spatial_model_1960$X0, spatial_model_1960$X0,
  theta = spatial_model_1960$theta) + diag(spatial_model_1960$g,
  nrow(spatial_model_1960$X0))
Kxs <- cov_gen(spatial_model_1960$X0, as.matrix(test_grid[,
  1]), theta = spatial_model_1960$theta)
Kss <- cov_gen(as.matrix(test_grid[, 1]), as.matrix(test_grid[,
  1]), theta = spatial_model_1960$theta)

Ki <- solve(Kx)
Sigma <- spatial_model_1960$nu_hat * (Kss - t(Kxs) %*%
  Ki %*% Kxs) + diag(spatial_1960$nugs)

# Simulated Null
Ynull1 <- MASS::mvrnorm(n = 2500, mu = prediction_1960,
  Sigma = Sigma)
Ynull2 <- MASS::mvrnorm(n = 2500, mu = prediction_1960,
  Sigma = Sigma)

resid <- Ynull1 - Ynull2
T0 <- colMeans(resid)
Tvar <- apply(resid, 2, var)
```



```

# Envelope statistics
R_i <- sort(apply(resid, 1, function(row) max(abs((row -
  T0)/sqrt(Tvar))))), decreasing = TRUE)
obs_diff <- prediction_1960 - prediction_1960

R_t <- max(abs((obs_diff - T0)/sqrt(Tvar))) # observed test statistic

alpha <- 0.05
q_index <- ceiling((1 - alpha) * length(R_i))

upper_env <- T0 + (R_i[q_index] * sqrt(Tvar))
lower_env <- T0 - (R_i[q_index] * sqrt(Tvar))

# Map envelope back to spatial limits
lat_mean <- prediction_1960 + T0
lat_upper <- prediction_1960 + upper_env
lat_lower <- prediction_1960 + lower_env

df_env <- data.frame(lon = x, lat_mean = lat_mean,
  lat_upper = lat_upper, lat_lower = lat_lower)

# Construct Polygon
poly_env <- data.frame(lon = c(x, rev(x), x[1]),
  lat = c(lat_lower, rev(lat_upper), lat_lower[1]))

poly_env_sf <- st_as_sf(poly_env, coords = c("lon",
  "lat"), crs = 4326) |>
  st_cast("POINT") |>
  st_combine() |>
  st_cast("POLYGON")

# Extract observation points for subsequent
# decades
other_decade <- subset(des_data, year > 1969)
points_sf <- st_as_sf(other_decade, coords = c("lon",
  "lat"), crs = 4326)

# Test if points fall inside polygon
# envelope
points_sf$inside_envelope <- st_within(points_sf,
  poly_env_sf, sparse = FALSE)[, 1]

```

Continental Representation of GET

Below is the Global Envelope Test for 1960 where shaded region are areas with the dashed blue line show the 95% confidence interval in the envelope, the red points indicate observations that fall outside envelope and the blue points fall within the envelope.

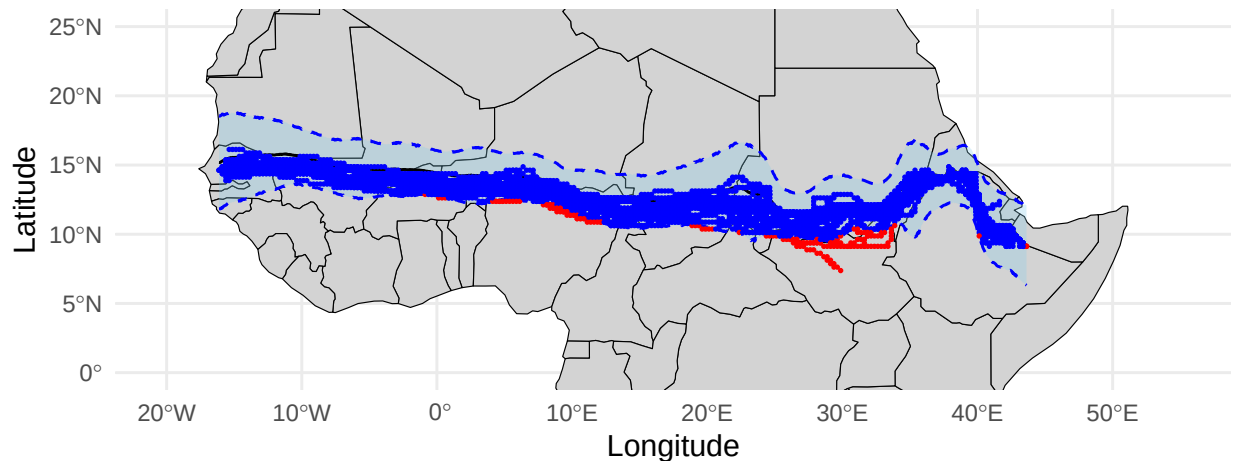
```

# Load Africa map
africa <- ne_countries(continent = "Africa", returnclass = "sf")

map_plot <- ggplot() + geom_sf(data = africa,
  fill = "lightgrey", color = "black") + geom_sf(data = poly_env_sf,
  fill = "skyblue", alpha = 0.3, color = NA) +
  geom_line(data = df_env, aes(x = lon, y = lat_upper),
    color = "blue", linetype = "dashed") +
  geom_line(data = df_env, aes(x = lon, y = lat_lower),
    color = "blue", linetype = "dashed") +
  geom_line(data = df_env, aes(x = lon, y = lat_mean),
    color = "black") + geom_sf(data = points_sf,
  aes(color = inside_envelope), size = 0.5) +
  scale_color_manual(values = c(`TRUE` = "blue",
    `FALSE` = "red")) + coord_sf(xlim = c(-20,
  55), ylim = c(0, 25)) + labs(x = "Longitude",
  y = "Latitude") + theme_minimal(base_size = 14) +
  theme(legend.position = "none")

# Print map
map_plot

```



Density of deviations by year

The histogram shows the percentage of which years have the highest amount of points that fall outside the 1960s GET. Note that the years 1983 and 1984 are lumped together given their proximity in time to each other.

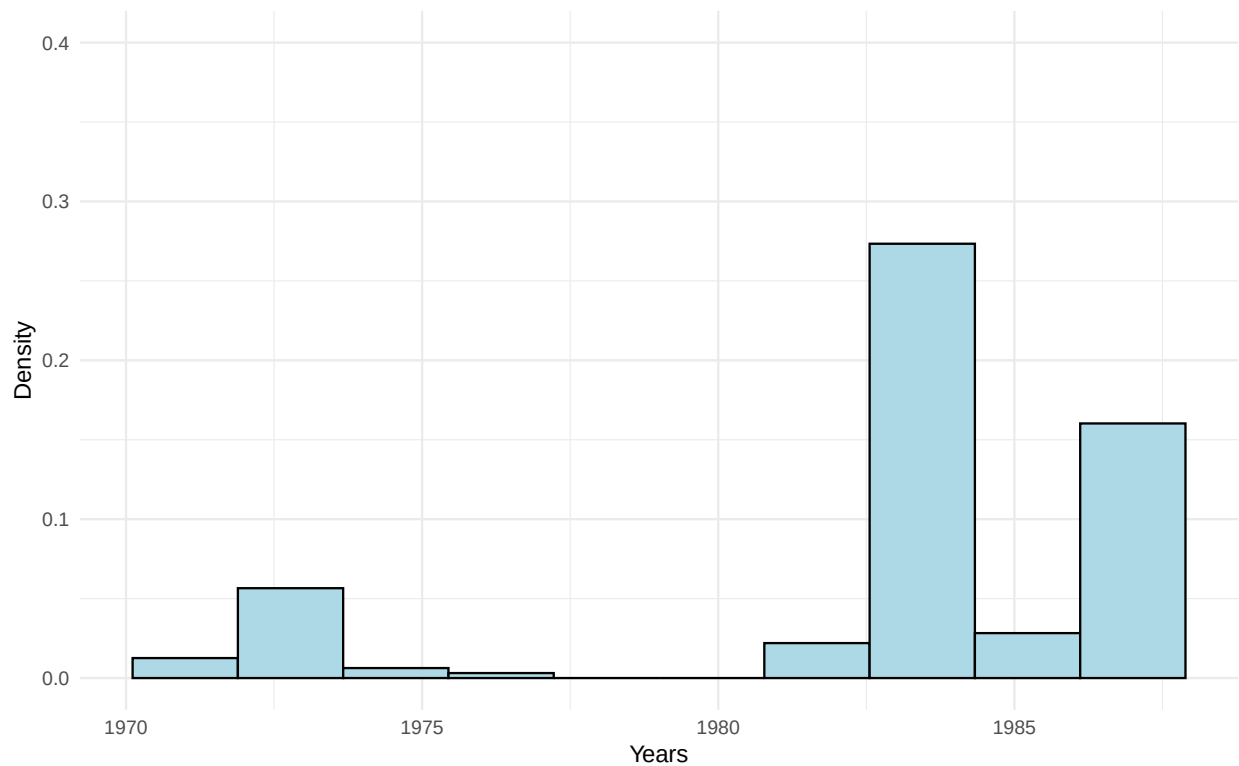
```
# Extract out-of-bounds years
Years <- subset(points_sf, inside_envelope ==
  FALSE)$year

hist_plot <- ggplot(data.frame(Years), aes(x = Years)) +
  geom_histogram(aes(y = after_stat(density)),
    bins = 10, fill = "lightblue", color = "black") +
  labs(x = "Years", y = "Density") + coord_cartesian(ylim = c(0,
    0.4)) + theme_minimal()

print(table(Years))
```

```
## Years
## 1971 1973 1974 1975 1977 1981 1982 1983 1984 1986 1987
##    4   18    1    1    1    1    6   45   42    9   51
```

```
# Print histogram
hist_plot
```



```
# Optional: Save Histogram
```

Conclusion

This outline provides a tangible format of how to implement temporal-spatial boundaries modeling while showing how an estimated boundary GET can be compared to boundary lines from year to year. Our flexible methodology can identify specific shifts or changes in space, while consider the historical narrative of

a particular region. Furthermore, we are able to apply a spatial testing methodology that can be translated to boundaries.